Supervised Learning for EDA
Logic Gate Representation Learning through GNN

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Z. Shi *et al.*, "DeepGate2: Functionality-Aware Circuit Representation Learning," *2023 IEEE/ACM International Conference on Computer Aided Design (ICCAD)*, San Francisco, CA, USA, 2023, pp. 1-9.

DeepGate2

- **Functionality-aware circuit representation learning**

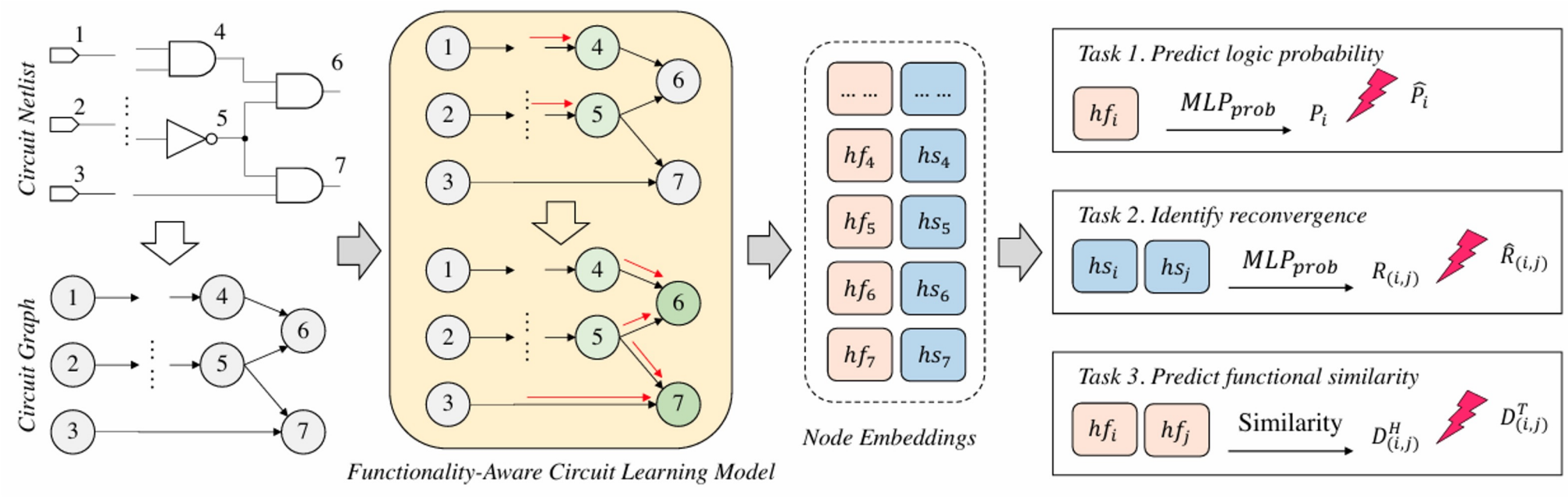

Fig. 4. Functionality-aware circuit learning framework

Download and Install DeepGate2

- Installation
  - conda create -n deepgate2 python=3.8.9
  - conda activate deepgate2
  - pip install -r requirements.txt (pytorch -gpu cause some problem)
  - Download .zip from github (https://github.com/zshi0616/DeepGate2)
  - cd dataset
  - tar -jxvf rawaig.tar.bz2
  - cd ..
  - python ./src/prepare_dataset.py --exp_id train --aig_folder ./dataset/rawaig # Use the default settings

- Refer to ./src/config.py for commands

Assignments

- **Assignment 1 (40%): Hidden layer dimensions**
  - Explore the dimensions of hidden layers to see how they impact the accuracy of Truth Table (TT) distance prediction and signal probability estimation
  - Experimental tip: Test dimensions like 32, 64, and 128
- **Assignment 2 (40%): Ablation study**
  - Conduct an ablation study to see the effectiveness of the pairwise Truth Table (TT) difference loss and the orthogonal PI (Primary Input) initialization
    - Pairwise TT loss: DeepGate2 introduced this to fix the "NOT gate problem" (where a gate and its inverse have the same signal probability but opposite functions). Removing this shows how much the model loses in functional similarity prediction.
    - Orthogonal PI initialization: DeepGate2 uses unique orthogonal vectors for each PI to mirror their functional independence. An ablation study would involve replacing these with homogeneous vectors to see how it affects reconvergence analysis.

Assignments

- **Assignment 3 (20%): Create a new task**

- **Brainstorming**
  - Switching probability prediction
    - Assume random patterns are applied
  - Redundant wire prediction
    - A redundant wire is a wire that can be replaced with a constant 0/1 without affecting the overall functionality
    - Wire representation learning? Or consider a wire is a fanin of a node
  - Roots of used cuts prediction in LUT mapping
  - …

Delivery & Due Date

- Delivery
  - To be announced

- Due on 2026/4/27 before class starts